

Achieving low write error rates in perpendicular spin orbit torque MRAM: A micromagnetic modeling feasibility study

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We present a micromagnetic modeling study of switching perpendicularly magnetized free layers with spin-orbit torques (SOTs) generated by spin currents consisting of only orthogonal (conventional SOT) as well as both orthogonal and collinear (unconventional SOT) spin polarizations relative to the free layer (FL) magnetization. Focusing on the FL diameter of 50 nm, for conventional SOT we find relatively high probabilities of FL back-hopping or incomplete switching at room temperature (T) for several bias configurations that deterministically switch the FL at $T = 0$ K. For unconventional SOT, fast, field-free deterministic switching is found at $T = 0$ K for orthogonal y -spin SOT efficiency $\theta_y = 0.3$ only when collinear z -spin efficiency $\theta_z/\theta_y > 0.25$. Such large ratio is needed to change the nature of the magnetization switching from a non-uniform state with dominant component along y to an almost fully reversed and uniform state along $-z$, resulting in thermally robust and complete switching without back-hopping.

Index Terms—spin transfer torque, SOT MRAM, perpendicular SOT, field-free switching, unconventional SOT, topological materials

I. INTRODUCTION

SPIN ORBIT torque magnetoresistive random access memory (SOT MRAM) has been proposed as an alternative to spin transfer torque (STT) MRAM in applications requiring fast memory with high endurance, such as low-level cache (LLC), where it is intended to replace SRAM [1]. In this regard, SOT MRAM where the free layer (FL) magnetization $\vec{M} = M\vec{z}$ is perpendicular to the plane (pSOT) has been extensively studied due to its potential for ultra-fast switching [2]–[3]. As shown in Fig. 1, when electrical current density J is applied through the SOT layer along $\vec{x}$, the spin polarization $\vec{\sigma}$ generated by the spin orbit coupling effects, such as the spin Hall effect, the Rashba-Edelstein effect, or the spin-momentum locking in topological insulators, is oriented along $\vec{y}$. As a consequence, damping-like component of SOT field, $\vec{H}_{SOT} \propto \vec{\sigma} \times \vec{M}$, has a non-zero initial value and can initiate switching instantaneously, instead of relying on thermal fluctuations to generate initial angle between $\vec{\sigma}$ and $\vec{M}$, causing incubation time delays as happens for in-plane SOT (or both in-plane and perpendicular STT devices when initially $\vec{\sigma} \parallel \vec{M}$ [1]). Such $\vec{H}_{SOT}$ tends to rotate $\vec{M}$ mostly in the xz plane instead of progressively increasing its angle with respect to $\vec{z}$ axis over several precessional cycles and can thus reverse $\vec{M}$ faster, on the order of sub-ns.

However, these advantages, both stemming from $\vec{\sigma}$ being orthogonal to $\vec{M}$, present this design with two fundamental challenges. Namely, when $\vec{\sigma} \parallel \vec{M}$, switching occurs by $\vec{H}_{SOT}$ acting directly against the damping torque, and thus the critical condition for switching is $\|\vec{H}_{SOT}\| = \alpha H_k$, where α is damping parameter and H_k is the net anisotropy field of the FL. On the contrary, when $\vec{\sigma} \perp \vec{M}$ the critical condition for switching is $\|\vec{H}_{SOT}\| = H_k/2$. As α is typically on the order of 0.01 this means that for pSOT the switching current densities are about two orders of magnitude higher compared to pSTT [4].

The other significant challenge arises from inherently non-deterministic nature of pSOT switching, as both up and down

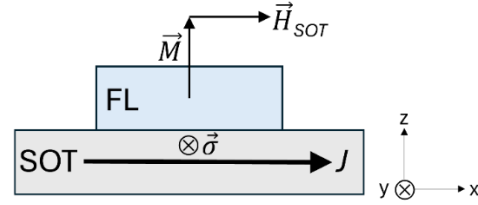


Fig. 1. A sketch of a perpendicular SOT device in its initial state, with spin orbit torque and free layers, current density J , spin current polarization $\vec{\sigma}$, free layer magnetization $\vec{M}$ and the effective SOT field $\vec{H}_{SOT}$.

$\vec{M}$ states are symmetric with respect to $\vec{\sigma}$. Various approaches have been proposed to break this symmetry [2] but none of them to date have demonstrated low error rate (≤ 1 ppm) writing, except in isolated, single device cases [5]–[6].

Recently, materials with low crystalline symmetry exhibiting unconventional SOT (USOT), where electrical current flowing along $\vec{x}$ generates spin polarizations along both $\vec{y}$ and $\vec{z}$ directions [7], have been proposed as a way to provide reliable deterministic switching. In this talk, we will report results of our micromagnetic modeling study of switching perpendicularly magnetized FLs with conventional as well as such USOTs in the domain wall reversal regime.

II. MODELING PARAMETERS

Micromagnetic modeling was performed using MuMax3, an open-source GPU-accelerated micromagnetic simulation program [8]. The FL parameters were diameter = 50 nm, thickness = 1.1 nm, $\alpha = 0.015$, exchange stiffness constant = 1.0 $\mu\text{erg/cm}$, saturation magnetization = 1000 emu/cm^3 and interfacial PMA energy density = 0.84 erg/cm^2 , resulting in $H_k = 4$ kOe. SOT efficiency for $\vec{y}$ spins was $\theta_y = 0.3$, while θ_z was varied from 0 to 0.5 relative to θ_y . USOT was modeled as a Slonczewski torque for $\vec{y}$ spins and a damping-like field term for $\vec{z}$ spins [9]–[10]. J pulse with vertical edge shape was applied at 1 ns after the start of each simulation and turned off at 3 ns. The simulations were stopped at 5 ns and were run at temperatures T of 0 K and 300 K.

III. RESULTS AND DISCUSSION

Figure 2 shows $T = 0$ K time traces of m_z , a normalized z component of $\vec{M}$ averaged over the FL, for $J_{min} = 1.79 \times 10^{12}$ A/m² (the minimum current density determined micromagnetically to rotate $\vec{M}$ to the xy plane for $\theta_z = 0$) and different ratio of SOT efficiencies θ_z/θ_y between 0 and 0.5. For conventional SOT case with $\theta_z = 0$, m_z reduces quickly from 1 to close to about zero (within 0.5 ns) and remains largely static until J pulse turns off. With increasing θ_z/θ_y to 0.25, m_z gradually increases in the negative direction, but starting at $\theta_z/\theta_y = 0.26$ there is a second transition to $m_z \cong -0.9$, which is faster for higher θ_z/θ_y . After J is turned off at $t = 3$ ns, m_z relaxes reliably within 2 ns to its fully reversed state for all $\theta_z/\theta_y \geq 0.26$, while for the lower θ_z/θ_y values the relaxation is much longer and could be unreliable at finite T . Fig. 2(b) shows that this transition corresponds to reorientation of $\vec{M}$ from a non-uniform state with dominant component along $-\hat{y}$ for $\theta_z/\theta_y = 0.25$, to an almost fully uniform state aligned mostly along $-\hat{z}$ for $\theta_z/\theta_y = 0.26$. This result also shows that relatively large values of $\theta_z/\theta_y > 0.25$ are needed for USOT to provide robust deterministic switching for the bit sizes of the order of 50 nm.

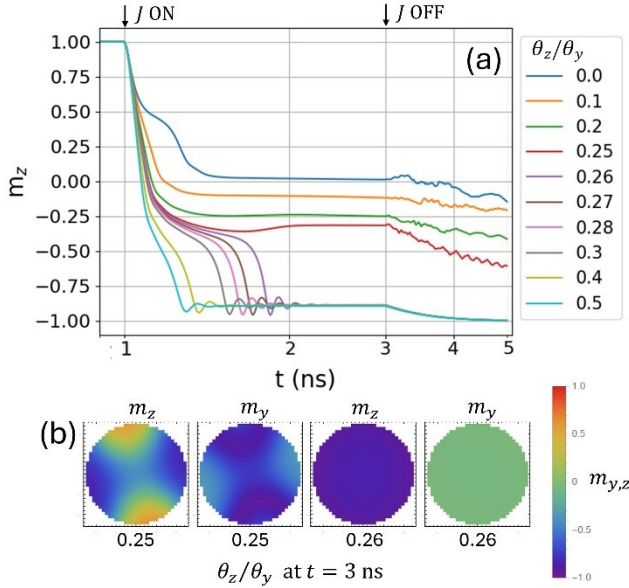


Fig. 2. (a) Time evolution of m_z as a function of θ_z/θ_y . (b) Spatial distribution of m_z and m_y across the FL taken at the end of the current pulse for $\theta_z/\theta_y = 0.25$ (left side) and 0.26 (right side).

Fig. 3 shows $T = 0$ K time trace of m_z for $\theta_z/\theta_y = 0.3$ as a function of J . We observe that the switching is still completed within 2 ns from the application of the current pulse for $J = 0.6 \times 10^{12}$ A/m² (see orange line in Fig. 3), a factor of 3 lower value compared to J_{min} despite somewhat decreasing switching speed. Reducing J further to 0.5×10^{12} A/m² results in incomplete switching (see blue line in Fig. 3).

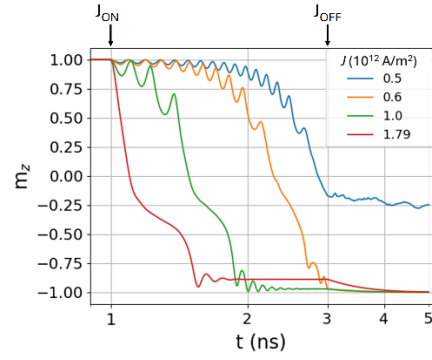


Fig. 3. Time evolution of m_z as a function of J for $\theta_z/\theta_y = 0.3$.

We also performed 100 simulations at $T = 300$ K for $J = 0.6 \times 10^{12}$ A/m², $\theta_z/\theta_y = 0.333$ and $\alpha = 0.015$ shown in Fig. 4. We obtained 100% switching without incomplete reversals at the end of the 2 ns long SOT pulse as well as no back-hopping at the end of each simulation (at $t = 5$ ns). Therefore, one can expect that unconventional SOTs should be able to provide field-free, fast and low WER operation for pSOT for $\theta_y \approx 0.3$ and $\theta_z/\theta_y \approx 0.3$ at room T .

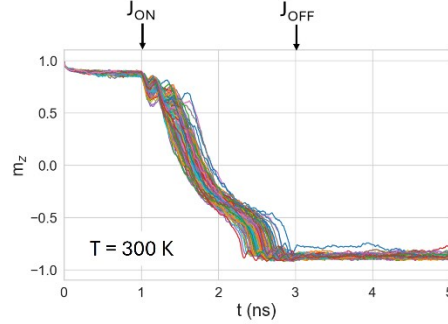


Fig. 4. $N = 100$ time evolutions of m_z for switching perpendicular FL with unconventional SOT at room T .

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